PRODUCT REQUIREMENTS DOCUMENT

Reducing Delivery Delay Impact on Power User Retention

Swiggy Food Delivery | Tier 1 Indian Cities | Peak Hour Experience

1. Background

Swiggy serves **24.3M monthly transacting users** and processed 294M festive-season orders (FY25, +26% YoY). Average delivery time: 37–38 minutes. However, Swiggy holds only **42–43% market share** vs Zomato's 55–58%. In quick commerce, Blinkit leads 40–45% vs Instamart's 20–25% with vastly better unit economics (-₹2 vs -₹18 per ₹100 GOV). CEO Sriharsha Majety committed to profitability by Q3 FY26, creating a tension: cost-cutting in logistics may be degrading the delivery experience that retains Swiggy's highest-value users.

2. Problem Statement

Working professionals in Tier 1 cities ordering during peak hours (12–2pm, 7–10pm) experience delivery times that regularly exceed the displayed ETA—with **72% of negative reviews** citing communication failure as the frustration amplifier. This affects power users (10–12 orders/month, ₹450 AOV) worth ₹14,850/year each. With **₹4.72 billion annual GOV at risk** from a 10% frequency drop, the cost of inaction is not hypothetical. Swiggy's ETA model doesn't proactively communicate delays or update expectations post-order. Meanwhile, Zomato invests in operational reliability with consistent 20–25 min windows. This PRD investigates whether ETA inaccuracy, partner shortages, or communication gaps are the primary driver—and defines what to build first.

2.1 Why Now

Zomato is investing in partner efficiency (faster assignment, route optimisation) while Swiggy's innovation focuses on Bolt (10-min delivery, 5–10% of orders), leaving 90%+ of standard deliveries underserved. 40% of negative Play Store reviews cite delays; 72% cite communication failure. Revenue at risk: 24.3M users × 18% power users × 10% at-risk × 2 fewer orders/month × ₹450 AOV × 12 months = **₹4,723 crore annually**. Losing one power user = 10–15× the acquisition cost.

3. Root Cause Analysis

The visible problem is simple: food arrived late. The causes run deeper. The iceberg model surfaces the hidden layers; the root cause table maps every specific failure point.

3.1 Iceberg Model

ABOVE THE SURFACE

VISIBLE EVENTS

Orders take 1hr+ instead of 30min. ETA shows 5min remaining for 30 more minutes. No one tells users why food is late. Cancellation charged despite Swiggy's failure.

RECURRING PATTERNS

Delays cluster in peak hours. ETA freezes at zero then goes silent. Partner assignment takes 30–40min after food is ready. Users who experience 2–3 delays quietly stop ordering.

---------- WATERLINE ----------

BELOW THE SURFACE

SYSTEMIC STRUCTURES
ETA model may not be recalibrated for peak-hour demand surges. Partners assigned multiple orders to cut cost-per-delivery. Notification system is reactive, not proactive. Support routed through chatbot with no human escalation.

UNDERLYING ASSUMPTIONS
"Profitability by Q3 FY26" means logistics cost cuts are acceptable. "Users care about speed" but data shows reliability matters more (0.78 vs 0.62 correlation). "Low checkout ETA drives conversion" — short-term gain, long-term trust erosion.

Key insight: We can't change Swiggy's profitability goals or fix partner supply overnight. But we **can** change what happens in the silence between the delay starting and the user losing trust. That's where this PRD intervenes.

3.2 Root Cause Table

Category	Fix Type	What Goes Wrong	How to Test
Restaurant-Side Delays	Ops + Product		
		Kitchen backlog during peak; inaccurate prep estimates; batching orders; slow confirmation causing reassignment lag	Compare peak vs off-peak prep times; measure estimated vs actual ready times; track time from order placed to confirmation
Not Enough Partners	Ops + Finance		
		Insufficient riders in busy zones; partners rejecting assignments; slow dispatch algorithm; incentive misalignment vs Zomato	Orders-per-rider ratio by zone/hour; acceptance rates peak vs off-peak; time from food ready to rider assigned
In-Transit Problems	Ops + Product		
		Suboptimal routing/traffic underestimation; multi-order batching hurting 2nd/3rd drop; rider detours and idle time	Planned vs actual GPS route time; single vs batched delivery time comparison; GPS deviation analysis
ETA Is Wrong	Product		
		Model not peak-aware; real-time signals not fed fast enough; ETA resets confusingly (countdown hits zero, jumps back up)	ETA error distribution peak vs off-peak; time from delay event to ETA update (target <60sec); frequency of ETA increases post-order
No Communication	Product		
		No proactive notification during delays; tracking shows a dot on map with no context; chatbot loops without resolution	% of delayed orders with notification sent; app open frequency during delays; support resolution rate vs ticket closure rate

The failure chain: Delay happens --> ETA lies about it --> Nobody tells the user --> User loses trust. The silence does more damage than the delay. Fixing communication breaks this chain even when the underlying delay persists.

Priority: (1) Communication gap [Product, 72% of reviews], (2) ETA accuracy [Product, 14 reviews], (3) Partner supply [Ops+Finance, 21 reviews]. This PRD specs solutions for #1 and #2. #3 is flagged for cross-functional action.

4. User Segmentation

Segment	Who	Orders	Cares About	When Late	Next Action
Silent Frequency Reducer	Finance analyst, Gurgaon. Daily grind.	15–20x/mo, ₹350–500	Value for money	Lunch overlaps meetings. Cold food. Stress compounds.	Silently reduces to 3x/week. No review. Monitors Zomato.
Churned Loyalist	IT manager, Bangalore. Young family.	12–15x/mo, ₹500–700	Reliability	Kids eat late. Midnight cooking.	Cancels Swiggy One. Switches to Zomato for months.
Premium User	Senior professional, Mumbai. Meetings.	8–10x/mo, ₹400–600	Speed	No lunch. Energy crashes.	Posts negative review. Halves orders. Trials Zomato Gold.
One-Timer	Freelancer, Delhi. Erratic schedule.	4–6x/mo, ₹300–450	Convenience	Work extends late.	Cooks next time. One-off review. No loyalty.

Primary target: Silent Frequency Reducer. Highest volume (15–20x/mo), invisible to current metrics (no reviews, no support tickets), most responsive to communication fixes. If 10% of estimated 2.2M silent reducers drop 2 orders/month, annual impact is ₹2,376 crore. The solution built for them also benefits the Churned Loyalist.

5. Data Assumptions

Metric	Estimate	Reasoning
Peak-hour orders exceeding ETA	35–45%	3–4M daily orders (Microsoft case study), 60% during peak. Swiggy's blog confirms riders are "far busier during Dinner Peak." 21/53 reviews cite partner delays.
Average delay beyond ETA	12–18 min	Review delays range 10min to 90min+; majority cluster 15–30min. Adjusting for negativity bias and Swiggy's "egregious delay" tracking category.
Delayed orders with proactive notification	5–15%	Zero of 53 reviewers mentioned receiving a notification. Swiggy's exception system exists but appears inactive during peak surges. Storyboard18 confirms support is "reactive."
Power users reducing frequency after 3+ delays	30–40%	ETA reliability correlates 0.78 with satisfaction vs 0.62 for speed. Low switching costs (Zomato one tap away). Silent reducers don't surface in complaints.
Daily delay-related support tickets	15K–22.5K	0.75% support incidence rate (Swiggy case study). Delivery complaints = 50–75% of all tickets during peak.
Support interactions resolving the issue	15–25%	72% of reviews cite support failure. Chatbot loops, templated responses. One user contacted support 8 times without resolution.

6. RICE Prioritisation

Score = (Reach x Impact x Confidence) / Effort. Top two selected for development.

Solution	Reach	Impact	Conf.	Effort	Score	Call	Rationale
Proactive delay notification	800K	3	90%	2	1.08M	BUILD	Addresses #1 pain point (72% cite silence). Infra exists. 2 person-months.
Peak-hour ETA buffer	2M	1	50%	1	1.00M	TEST	High score but 50% confidence. Could reduce checkout conversion. Needs A/B test first.
Live transparency screen	800K	2	80%	3	427K	BUILD	Pairs with #1: notification tells you, screen shows you. Closes communication gap.
Post-delay recovery flow	800K	1	80%	2	320K	LATER	Reactive. If #1 and #3 work, fewer users need recovery.
Dynamic ETA recalculation	800K	2	70%	4	280K	DEFER	Highest effort, lowest confidence. Transparency screen achieves similar trust effect.

Decision: Build Solution 1 (notification) + Solution 3 (transparency screen). Combined: 5 person-months. Together they close the communication gap without requiring ops or ETA model changes.

7. Product Specification

Rahul Kapoor

Silent Frequency Reducer | PRIMARY

27, Finance Analyst, Gurgaon. 15–20x/mo, ₹350–500.

"I don't complain. I just order less."

Meera Iyer

Churned Loyalist | SECONDARY

34, IT Manager, Bangalore. 12–15x/mo, ₹500–700.

"If my kids eat late again, I'm done."

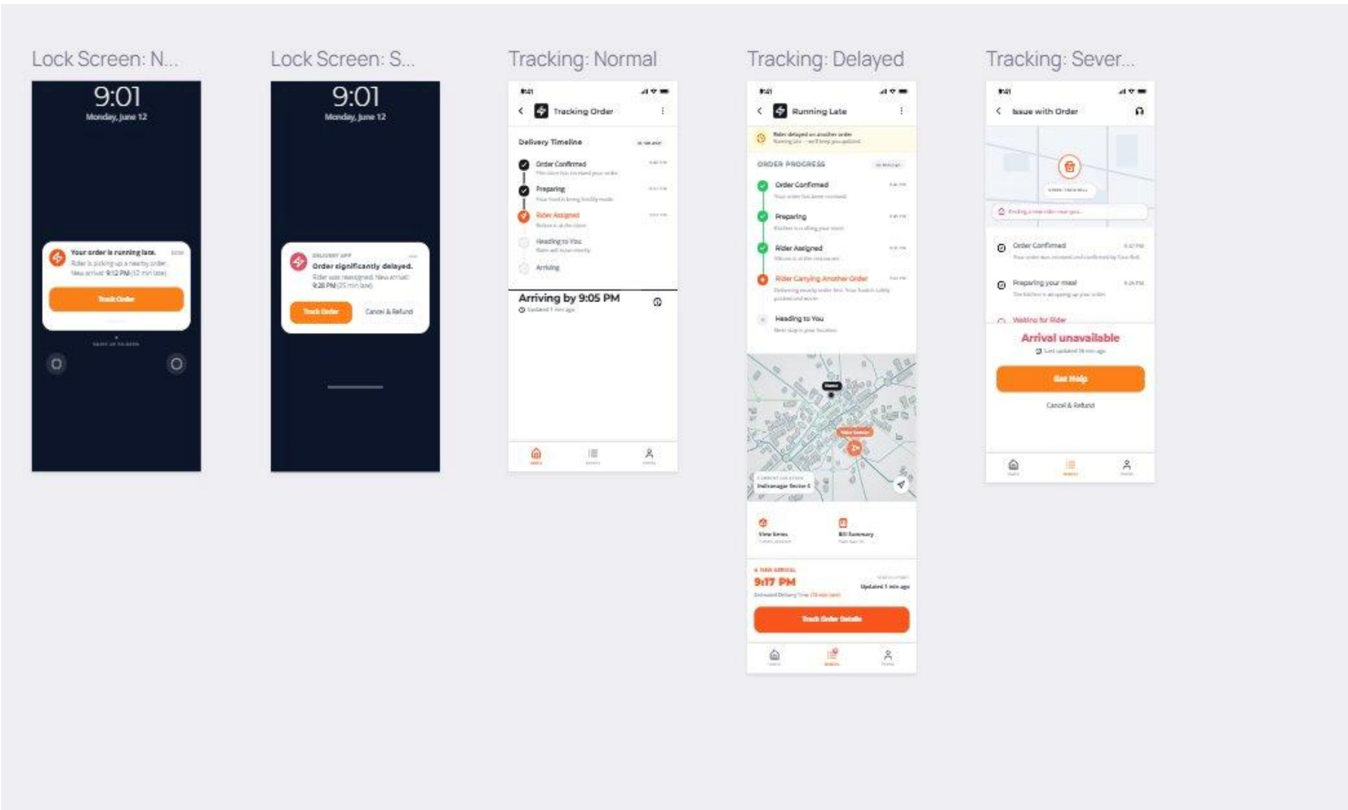
	Solution 1: Proactive Delay Notification	Solution 2: Live Transparency Screen
What it does	Push notification when system detects 8+ min delay, with reason and updated ETA—before the user notices.	Stage-by-stage timeline replacing the dot-on-map, showing exactly where the order is and what's causing hold-ups.
User stories	Rahul: Notify me the moment my order might be late so I can grab a snack. Meera: Tell me the real arrival time so I can decide to cancel and cook.	Rahul: Show me clear stage labels so I know where the hold-up is. Meera: Tell me when the rider is delivering someone else's order first.
Acceptance criteria	Fires within 30 sec of 8+ min delay detection. Includes reason + updated ETA + Track Order link. At 20+ min: one-tap Cancel & Refund. Max 2 notifications per order.	Updates within 5 sec of stage change. 6 stages: Preparing, Waiting, Assigned, Carrying Another, Heading, Arriving. Delayed stage shows context message. Get Help button at 15+ min stuck (human agent).
Edge cases	2nd delay: Revised ETA, not generic repeat. GPS loss 3+ min: "Rider temporarily offline." Notifications off: in-app banner. False positive resolves: stay silent.	Restaurant stalling 10+ min: "high volume" message. Rider moving away: "fulfilling another order." Stage unknown: "Updating status..." Rider swap: show new name + reset ETA.
Dependencies	Data Science: 75%+ prediction accuracy. Rider App: real-time status <10 sec. Infra: existing FCM/APNs.	Frontend: WebSocket, iOS + Android. Restaurant Ops: 80%+ Order Ready compliance. Backend: expand API to 6 stages.
Effort	2 person-months	3 person-months

Rollout: Solution 1 builds weeks 1–4, pilots week 5–6 at 10% in Bangalore. Solution 2 builds weeks 5–10, pilots week 11–12. Both live Tier 1 by week 13. Go/no-go: false positive rate <25%, notification open rate >40%.

8. Wireframes

#	Screen	Solution	What It Shows
1	Lock Screen: Normal Delay	Notification	Push notification at 8+ min delay. Reason, updated ETA, Track Order tap target.
2	Lock Screen: Severe Delay	Notification	20+ min delay. Adds Cancel & Refund alongside Track Order.
3	Tracking: Normal	Transparency	6-stage timeline, active stage in orange, checkmarks for completed. Map + large ETA.
4	Tracking: Delayed	Transparency	"Rider Carrying Another Order" active. Context message. Rider moving away on map.
5	Tracking: Severe	Both	Stuck 15+ min. "Arrival unavailable." Get Help button + Cancel & Refund.

User flow: Delay detected --> Notification (Screen 1/2) --> Tap Track Order --> Delay tracking (Screen 4) --> If worsens --> Severe state with Get Help + Cancel (Screen 5)



Prototype link: [▶ Swiggy Delay Notification Wireframes]

9. Metrics Framework

Unlike the impact tree in Section 3 (which traced causes to damage), this traces our intervention forward: what we ship leads to what behaviour change, which drives what business result.

NORTH STAR: Repeat order rate among delayed-order users within 14 days
Baseline: ~45% | Target: 60%+ | Timeframe: 90 days post full launch

What We Ship		Behaviour Change		Business Outcome
Notification sent within 30 sec	-->	Users feel informed before noticing delay	-->	Fewer support tickets for delayed orders
Notification includes reason + updated ETA	-->	Users trust the updated time and wait	-->	Lower cancellation rate for delayed orders
Tracking shows clear stage labels + context	-->	Users stop anxiously reopening app	-->	Higher repeat order rate post-delay
Get Help at 15+ min connects to human	-->	Severe delays get real resolution	-->	NPS improvement for delayed-order users
Cancel & Refund at 20+ min, one tap	-->	Users feel control, not resentment	-->	Fewer 1-star reviews mentioning delays

9.1 Leading Indicators

Metric	Baseline	Target	Why It Matters
Notification open rate	--	>40%	Below 30% = content not compelling or notifications disabled.
False positive rate	--	<25%	Above 30% erodes trust in the notification system itself.
Support tickets (delayed orders)	~18K/day	-30%	Proactive info should reduce need to contact support.
Tracking session duration (delays)	~4.5 min	<2 min	Long sessions = anxious checking. Shorter = got the info needed.
Cancellation rate (10+ min delays)	~18%	<12%	If users trust updated ETA, they wait instead of cancelling.
Get Help button usage	--	<5%	Low = transparency screen provides enough clarity.
Repeat order rate (14 days)	~45%	60%+	The north star. Do delayed-order users come back?

Week 5–6 (pilot): Monitor notification open rate and false positive rate daily. Pause if false positives >30%.

Week 7–8 (expansion): Track support ticket reduction and cancellation rate weekly. Compare pilot vs control.

Week 13+ (full launch): Measure north star monthly. Report to leadership quarterly alongside GOV impact.

Kill criteria: No north star improvement after 60 days of full launch = escalate. Consider pivoting to dynamic ETA recalculation.